8 AI for Industrial Knowledge Intelligence: Unified Asset & Operations Brain

Theme: *Industrial Intelligence / Document Management / Knowledge Engineering / Quality*

PROBLEM CONTEXT

A 2024 McKinsey global survey found that professionals in asset-intensive industries spend an average of 35% of their working hours searching for information, clarifying instructions, or recreating documents that already exist somewhere in the organisation. In India specifically, a NASSCOM-EY study of manufacturing and energy companies found that the average large plant operates across 7 to 12 disconnected document systems — P&IDs and engineering drawings in one place, maintenance work orders in another, operating procedures in a third, inspection records in a fourth, and regulatory submissions scattered across email archives. BIS Research estimated that this fragmentation contributes to 18–22% of unplanned downtime events in Indian heavy industry, as maintenance teams make decisions without complete equipment history or failure pattern context. Then there is the knowledge cliff: an estimated 25% of India's experienced industrial engineers and operators will retire within the next decade, taking decades of undocumented operational knowledge with them. Once gone, it cannot be recovered. Knowledge fragmentation in industrial operations is not a file management problem. It is a safety problem, a quality problem, and an operational efficiency problem — and it compounds over time. The organisations that solve it first will have a structural advantage in how they operate, maintain, and improve their assets.

CHALLENGE STATEMENT

Build an AI-powered Industrial Knowledge Intelligence platform that ingests heterogeneous documents — engineering drawings, maintenance records, safety procedures, inspection reports, operating instructions, project files — across structured and unstructured formats, and makes their collective intelligence queryable, actionable, and continuously updated at the point of need, across any device or function.

WHAT YOU MAY BUILD

Participants may explore areas such as:

- **Universal Document Ingestion & Knowledge Graph Agent —** AI pipeline that processes PDFs, P&IDs, scanned forms, spreadsheets, and email archives — extracting entities (equipment tags, process parameters, regulatory references, personnel, dates) and building a unified knowledge graph that maintains relationships across document types and updates automatically as new records arrive.
- **Expert Knowledge Copilot —** RAG-powered conversational AI that answers operational, maintenance, and engineering queries across the full document corpus — with source citations, confidence scores, and direct links to the originating documents. Built to work on mobile for field technicians, not just desktops for engineers.
- **Maintenance Intelligence & RCA Agent —** AI agent that fuses work order history, equipment failure records, OEM manuals, inspection findings, and real-time operating conditions to generate predictive maintenance recommendations, Root Cause Analysis (RCA) support, and optimised maintenance schedules — reducing unplanned downtime by connecting the dots that no individual team member can connect alone.
- **Quality & Regulatory Compliance Intelligence —** Agentic system that maps regulatory requirements (Factory Act, OISD, PESO, environmental norms, quality standards) against current procedures, equipment states, and inspection records — identifying compliance

gaps, auto-generating compliance evidence packages for audits, and flagging quality deviations before they escalate.
- **Lessons Learned & Failure Intelligence Engine —** AI agent that analyses incident reports, near-miss records, audit findings, and quality non-conformances across the organisation's history and external industry databases — identifying systemic patterns invisible to any individual review, and proactively pushing relevant warnings to operational teams before similar conditions recur.

These examples are illustrative only.

SUGGESTED TECHNOLOGIES

- RAG (Retrieval-Augmented Generation) over heterogeneous industrial document corpora
- Knowledge Graphs & Industrial Ontology Engineering
- Computer Vision (P&ID parsing, drawing digitisation)
- OCR & Document Intelligence (structured + unstructured)
- Quality Management System (QMS) Integration
- Agentic AI for maintenance and compliance workflows

EXPECTED DELIVERABLES

- Working Prototype
- Architecture Diagram
- Presentation Deck
- Demo Video

Evaluation Focus Entity extraction accuracy across document types, query answer quality on domain-expert benchmark questions, knowledge graph linkage completeness, time-to-answer versus traditional search, compliance gap detection accuracy, and demonstrated improvement in cross-functional knowledge discovery — ideally validated with real industrial document samples.

JUDGING CRITERIA

Criteria	Weight
Innovation	25%
Business Impact	25%
Technical Excellence	20%
Scalability	15%
User Experience	15%